A First Course in Abstract Algebra

Section 5: Subgroups

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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5 Subgroups

Exercise 5.1: Real numbers in the additive complex group

Yes. The real numbers contain 0, and for any reals a, b both $a + b$ and $-a$ are real. Associativity is inherited from complex addition. These facts verify the subgroup conditions.

Exercise 5.2: Positive rationals additively

No. The additive identity 0 is not positive, so it is not in $\mathbb{Q}^+$. Also a positive rational has a negative additive inverse, outside the set. Closure under addition alone is insufficient.

Exercise 5.3: Multiples of seven

Yes. The set $7\mathbb{Z}$ contains zero. If $7m, 7n \in 7\mathbb{Z}$, then $7m + 7n = 7(m + n)$ and $-(7m) = 7(-m)$ remain in the set. Thus it is an additive subgroup of $\mathbb{C}$.

Exercise 5.4: Pure imaginary numbers

Yes. The identity is $0 = i \cdot 0$. For real a, b, $ia + ib = i(a + b)$ and $-(ia) = i(-a)$ are pure imaginary. Therefore closure and inverse closure hold.

Exercise 5.5: Rational multiples of pi

Yes. The set is $\pi\mathbb{Q} = \{\pi q : q \in \mathbb{Q}\}$. It contains zero, sums have the form $\pi(q + r)$, and negatives have the form $\pi(-q)$. Since $q + r, -q$ are rational, all subgroup conditions hold under addition.

Exercise 5.6: Integral powers of pi additively

No. Every π^n is positive, so zero is absent and no negative additive inverse is included. Thus $\{\pi^n : n \in \mathbb{Z}\}$ is not an additive subgroup, regardless of whether particular sums happen to lie in it.

Exercise 5.7: The same six sets multiplicatively

The only multiplicative subgroups of $\mathbb{C}^*$ among the six are $\mathbb{Q}^+$ and $\{\pi^n : n \in \mathbb{Z}\}$. The positive rationals contain 1, are closed under products, and have positive rational reciprocals. Integral powers of π contain $\pi^0 = 1$, multiply by adding exponents, and invert by negating exponents. The other four sets $(\mathbb{R}, 7\mathbb{Z}, i\mathbb{R}, \pi\mathbb{Q})$ contain 0, so are not even subsets of the stated group $\mathbb{C}^*$.

Exercise 5.8: Matrices of determinant two

No. The identity matrix has determinant 1, so it is absent. In addition, the product of two matrices of determinant 2 has determinant 4, violating closure. Such matrices exist, for example $\mathrm{diag}(2, 1, \ldots, 1)$.

Exercise 5.9: Invertible diagonal matrices

Yes. The identity is diagonal, products multiply corresponding nonzero diagonal entries, and reciprocating those entries produces a diagonal inverse. Thus this nonempty subset is closed under products and inverses, hence is a subgroup of $GL(n, \mathbb{R})$.

Exercise 5.10: Invertible upper-triangular matrices

Yes. Products remain upper triangular and their diagonal entries are the corresponding products of nonzero diagonal entries. The identity belongs. Back substitution in $Ax = e_j$ shows A^{-1} is upper triangular, and its diagonal entries are $1/a_{jj}$, hence nonzero. Thus this is a subgroup.

Exercise 5.11: Matrices of determinant minus one

No. The identity has determinant 1, and the product of two determinant--1 matrices has determinant 1, outside the set. Hence both identity membership and product closure fail.

Exercise 5.12: Matrices of determinant one or minus one

Yes. The identity belongs, products have determinant in $\{1, -1\}$, and the inverse has determinant $1/\det A$, again in $\{1, -1\}$. All matrices in the set are invertible, so the subgroup criterion applies.

Exercise 5.13: Orthogonal matrices

The identity satisfies $I_n^T I_n = I_n$. If $A^T A = B^T B = I_n$, then $(AB)^T(AB) = B^T A^T AB = B^T B = I_n$, so products remain orthogonal. The equation $A^T A = I_n$ gives $A^{-1} = A^T$, and then $(A^{-1})^T A^{-1} = AA^T = I_n$. Thus inverses also belong, and this is a subgroup.

Exercise 5.14: Nowhere-zero functions

(a) Not an additive subgroup: the zero function is absent; also $1 + (-1) = 0$ shows failure of closure. **(b)** Yes, this is the entire multiplicative group $\widetilde{F}$. Products are nowhere zero, the constant 1 is the identity, and the inverse of f is the function $x \mapsto 1/f(x)$, defined everywhere because f never vanishes.

Exercise 5.15: Functions vanishing at one

(a) Yes. The zero function belongs, and if $f(1) = g(1) = 0$, then $(f + g)(1) = 0$ and $(-f)(1) = 0$. **(b)** No. Every function in the specified set vanishes at 1, so none belongs to $\widetilde{F}$. It is not a subset of the proposed ambient multiplicative group.

Exercise 5.16: Nowhere-zero functions equal to one at one

(a) No: the zero function is absent (and adding two members gives value 2 at 1). **(b)** Yes. The constant function 1 belongs, products have value $1 \cdot 1 = 1$ at 1, and reciprocals have value $1/1 = 1$

there. Nowhere-zero values are preserved by products and reciprocals.

Exercise 5.17: Nowhere-zero functions equal to one at zero

(a) No, since the additive identity is absent and adding two such functions gives value 2 at zero. **(b)** Yes: the constant 1 belongs, and both $(fg)(0) = f(0)g(0) = 1$ and $(1/f)(0) = 1$ hold. All resulting functions remain nowhere zero.

Exercise 5.18: Nowhere-zero functions equal to minus one at zero

(a) No: the zero function does not satisfy the defining condition. **(b)** No: the multiplicative identity has value 1, not -1, at zero. Furthermore multiplying two members gives value $(-1)(-1) = 1$ at zero, so product closure fails.

Exercise 5.19: Nonzero constant functions

(a) No: the zero constant is not in this subset of $\widetilde{F}$, and the constants $1, -1$ add to it. **(b)** Yes: nonzero constants contain 1, multiply to nonzero constants, and have nonzero constant reciprocals. Thus they form a multiplicative subgroup.

Exercise 5.20: All subgroup relations among nine groups

Include the nine reflexive relations $G_i \leq G_i$. All strict relations are the transitive consequences of

$$G_2 < G_8 < G_7 < G_1 < G_4, \qquad G_9 < G_3 < G_5, \qquad G_6 < G_5.$$

Explicitly the additional pairs (i, j) with $G_i < G_j$ are

$$(2,8),(2,7),(2,1),(2,4),(8,7),(8,1),(8,4),$$
$$(7,1),(7,4),(1,4),(9,3),(9,5),(3,5),(6,5).$$

The additive chain is $12\mathbb{Z} \subset 6\mathbb{Z} \subset 3\mathbb{Z} \subset \mathbb{Z} \subset \mathbb{R}$. Multiplicatively, powers of 6 are positive rationals, and powers of π are positive reals. The latter are not all rational, since π is irrational; the two cyclic power groups do not contain each other, since π is not a power of 6 and 6 is not an integral power of π (use $\pi < 6 < \pi^2$). A subgroup must use the ambient operation, so no cross-relations between the additive and multiplicative lists are allowed.

Exercise 5.21: Five elements of three cyclic groups

(a) In $25\mathbb{Z}$: $-50, -25, 0, 25, 50$, obtained using integer multiples $-2, -1, 0, 1, 2$ of 25.

(b) In $\{(1/2)^n : n \in \mathbb{Z}\}$: $4, 2, 1, 1/2, 1/4$, from those same exponents.

(c) In $\{\pi^n : n \in \mathbb{Z}\}$: $\pi^{-2}, \pi^{-1}, 1, \pi, \pi^2$.

Zero exponents in a multiplicative cyclic group give its identity 1; they never give the number 0.

Exercise 5.22: A signed interchange matrix

Let $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. Direct multiplication gives $A^2 = I_2$, while $A \neq I_2$. Thus every even integral power is I_2 and every odd integral power is A (also for negative exponents, since $A^{-1} = A$). The subgroup is precisely $\{I_2, A\}$.

Exercise 5.23: A shear matrix

For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, multiplication shows

$$\begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & m+n \\ 0 & 1 \end{pmatrix}.$$

Thus induction gives $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for positive n, and the inverse $A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ extends it to every integer n. These, and only these, are the elements of $\langle A \rangle$.

Exercise 5.24: A diagonal generator

If $A = \operatorname{diag}(3, 2)$, then $A^n = \operatorname{diag}(3^n, 2^n)$ for every $n \in \mathbb{Z}$; negative powers use reciprocals of the diagonal entries. Therefore

$$\langle A \rangle = \{\operatorname{diag}(3^n, 2^n) : n \in \mathbb{Z}\}.$$

The same exponent must occur in both positions; the subgroup does not allow them to be chosen independently.

Exercise 5.25: A scaled signed interchange

Write $B = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = 2A$ with $A^2 = I_2$ from Exercise 5.22. Since scalar matrices commute with A, $B^n = 2^n A^n$. Thus its elements are

$$\{2^{2k} I_2 : k \in \mathbb{Z}\} \cup \left\{ 2^{2k+1} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} : k \in \mathbb{Z} \right\}.$$

This includes negative powers; $B^2 = 4I_2$ shows that the group is infinite.

Exercise 5.26: Which six groups are cyclic?

G_1 : Cyclic, generated exactly by $1, -1$: the multiples of an integer a fill $\mathbb{Z}$ only if $|a| = 1$.

G_2 : Not cyclic. If $q \neq 0$ were an additive generator, $q/2$ could not be an integral multiple of q; $q = 0$ plainly cannot generate $\mathbb{Q}$.

G_3 : Not cyclic. If a positive rational generator differs from 1, replace it by its reciprocal if necessary so it is $q > 1$. Then the rational $(1 + q)/2$ lies strictly between 1 and q, while no integral power of q does. The generator 1 also fails.

G_4 : Cyclic, with precisely the generators $6, -6$, since division by 6 identifies this group with $\mathbb{Z}$.

G_5 : Cyclic, with precisely $6, 1/6$ as generators. If 6^k generates, there must be an integer m with $6^{km} = 6$, so $km = 1$ and $k = \pm 1$.

G_6 : Not cyclic. If $a + b\sqrt{2}$ generated it, expressing $1 = k(a + b\sqrt{2})$ with integers forces $kb = 0$ and $ka = 1$, by irrationality of $\sqrt{2}$. Thus $b = 0$, $a = \pm 1$, and its multiples cannot include $\sqrt{2}$.

Exercise 5.27: The subgroup generated by three modulo four

Successive multiples of 3 modulo 4 are $0, 3, 2, 1$, then 0 again. These four residues are distinct, so the generated subgroup has order 4.

Exercise 5.28: A nonidentity element of the Klein group

Table 5.11 gives $c^2 = e$ and $c \neq e$. Therefore the subgroup generated by c is $\{e, c\}$ and has order 2.

Exercise 5.29: An angle of two pi over three

The generator is $e^{2\pi i/3}$. Its kth power is 1 exactly when $2\pi k/3$ is an integral multiple of 2π, or $3 \mid k$. The least positive such k is 3, so the subgroup has order 3, even though the ambient group has six elements.

Exercise 5.30: An angle of four pi over five

The generator is $e^{4\pi i/5}$. Its kth power is 1 if and only if $5 \mid 2k$. Since 2 and 5 are relatively prime, this is equivalent to $5 \mid k$. Thus the order is 5.

Exercise 5.31: An angle of three pi over two

The generator is $e^{3\pi i/2} = -i$. Its powers are $1, -i, -1, i$, then 1 again. Consequently its cyclic subgroup has order 4.

Exercise 5.32: An angle of five pi over four

The generator is $e^{5\pi i/4} = e^{2\pi i(5/8)}$. Its kth power is 1 exactly when $8 \mid 5k$. Because $\gcd(5, 8) = 1$, the least positive such k is 8. Thus this element generates all of U_8.

Exercise 5.33: A permutation matrix of order two

On the standard column vectors the matrix interchanges e_1, e_3 and interchanges e_2, e_4. Applying it twice fixes all four basis vectors, so $A^2 = I_4$. Since one application does not fix e_1, $A \neq I_4$. Therefore

its cyclic subgroup has order 2.

Exercise 5.34: A permutation matrix of order four

Track the first standard basis vector: $e_1 \mapsto e_3 \mapsto e_2 \mapsto e_4 \mapsto e_1$. This is a cycle containing all four basis vectors. Four applications fix each vector, and fewer than four do not fix e_1. Thus $A^4 = I_4$ with no smaller positive identity power, giving order 4.

Exercise 5.35: A permutation matrix of order three

This matrix sends $e_1 \mapsto e_4 \mapsto e_2 \mapsto e_1$ and fixes e_3. Thus three applications fix every standard basis vector. Neither one nor two applications fix e_1, so the least positive identity power is 3, and the generated subgroup has order 3.

Exercise 5.36: All subgroups generated in Z six

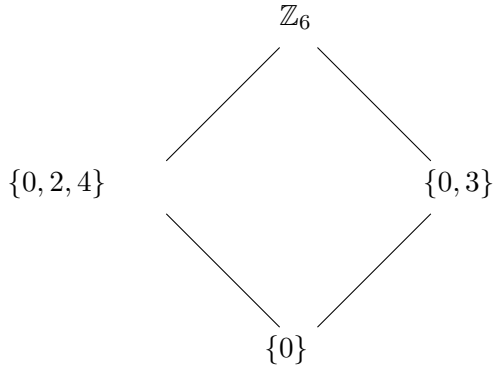
(a) Add row and column labels modulo 6:

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(b) $\langle 0 \rangle = \{0\}$, $\langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6$, $\langle 2 \rangle = \langle 4 \rangle = \{0, 2, 4\}$, $\langle 3 \rangle = \{0, 3\}$. These follow by repeatedly adding the generator until zero reappears.

(c) Exactly 1 and 5 generate all six residues.

(d) The inclusion diagram is the diamond below. Neither middle subgroup contains the other, since $2 \notin \{0, 3\}$ and $3 \notin \{0, 2, 4\}$.



Exercise 5.37: A complete subgroup definition

A subgroup H of a group G is a subset that is itself a group under the operation inherited from G. Equivalently, it contains e, is closed under products, and contains the inverse of each of its elements. The proposed definition omits product closure. For instance $\{0, 1, -1\}$ in the additive group $\mathbb{Z}$ contains the identity and all negatives but is not closed under addition.

Exercise 5.38: The definition of cyclic

The stated definition is correct: there exists $a \in G$ such that every element of G is a^n for some integer n. Both negative powers and $n = 0$ are allowed. In additive notation the same condition is $G = \{na : n \in \mathbb{Z}\}$.

Exercise 5.39: Ten subgroup statements

(a) True: associativity is one of the group axioms.

(b) False: multiplication by inverses proves both cancellation laws in every group.

(c) True: the inherited operation makes G a subgroup of itself.

(d) False: the text calls only G itself the improper subgroup; $\{e\}$ is the trivial subgroup, usually proper when $G \neq \{e\}$.

(e) False: in $\mathbb{Z}_4$, the element 2 generates only $\{0, 2\}$.

(f) False: $1, -1$ are distinct generators of the infinite cyclic group $\mathbb{Z}$.

(g) False: $\mathbb{Z}$ is a group under addition but zero has no multiplicative inverse.

(h) False as a proposed complete definition: a subgroup is a subset satisfying additional operation and group conditions.

(i) True: 1 generates $\mathbb{Z}_4$ by addition modulo 4.

(j) False: $\{1\}$ is a subset of the additive group $\mathbb{Z}$ but is not a subgroup, since it lacks zero.

Exercise 5.40: More than two square roots of the identity

In the Klein four-group $V = \{e, a, b, c\}$ every element satisfies $x^2 = e$. Thus there are four solutions, exceeding two. The familiar degree bound for polynomial roots over a field cannot be applied to an arbitrary group equation.

Exercise 5.41: An isomorphism carries subgroups to subgroups

Since H contains e, its image contains $\phi(e) = e'$. If $u = \phi(h)$ and $v = \phi(k)$ lie in $\phi[H]$, then $u *' v = \phi(h * k)$ belongs to $\phi[H]$ because $h * k \in H$. Also $u^{-1} = \phi(h^{-1})$ belongs because $h^{-1} \in H$. Thus the image contains the identity and is closed under products and inverses, so is a subgroup of G'.

Exercise 5.42: An isomorphism carries a cyclic group to a cyclic group

Let $G = \langle a \rangle$. Every $g' \in G'$ is $\phi(g)$ for some $g \in G$, by surjectivity. Write $g = a^n$. A homomorphism preserves positive powers by repeated multiplication, zero powers because it preserves the identity, and negative powers because it preserves inverses. Hence $g' = \phi(a)^n$. Thus $G' = \langle \phi(a) \rangle$ is cyclic.

Exercise 5.43: Products of two subgroups in an abelian group

Let $HK = \{hk : h \in H, k \in K\}$. It contains $ee = e$. If $h_1k_1, h_2k_2 \in HK$, then by commutativity

$$(h_1k_1)(h_2k_2) = (h_1h_2)(k_1k_2) \in HK.$$

Also $(hk)^{-1} = k^{-1}h^{-1} = h^{-1}k^{-1} \in HK$. The factors remain in their respective subgroups. Therefore HK is a subgroup; commutativity was essential in regrouping the factors.

Exercise 5.44: The hidden nonemptiness assumption

The argument starts "let $a \in H$," which is not possible when $H = \varnothing$. The empty subset is vacuously closed under products and inverses, so Conditions 1 and 3 alone do not force identity membership. If one separately assumes H nonempty, the argument becomes valid. The flaw is therefore not the algebraic calculation, but the unjustified choice of an element.

Exercise 5.45: The one-step subgroup criterion

If H is a subgroup, then $b^{-1} \in H$ and hence $ab^{-1} \in H$ for $a, b \in H$. Conversely assume $H \neq \varnothing$ and that all such products belong. Choose $h \in H$; then $hh^{-1} = e \in H$. For $b \in H$ use $a = e$ to get $b^{-1} \in H$. Finally for $a, b \in H$ the element b^{-1} also belongs, so $a(b^{-1})^{-1} = ab \in H$. Thus identity, inverse closure, and product closure all follow.

Exercise 5.46: Only one generator

If $G = \langle a \rangle$, then a^{-1} is also a generator because $(a^{-1})^{-n} = a^n$. With only one generator we must have $a = a^{-1}$, so $a^2 = e$. Every power of a is then either e or a, giving $|G| \leq 2$. This includes the trivial group, whose sole generator is its identity.

Exercise 5.47: The solutions of x squared equals identity

Let $H = \{x : x^2 = e\}$ in an abelian group. The identity belongs. If $x, y \in H$, commutativity gives $(xy)^2 = x^2y^2 = e$. Also $(x^{-1})^2 = (x^2)^{-1} = e$. Thus H is closed under products and inverses and is a subgroup. Without commutativity, the regrouping of $(xy)^2$ would not be justified.

Exercise 5.48: The solutions of a fixed power equation

For fixed $n \geq 1$, put $H = \{x : x^n = e\}$. The identity is in H. If $x, y \in H$, the commuting-product power law gives $(xy)^n = x^ny^n = e$, and $(x^{-1})^n = (x^n)^{-1} = e$. Hence the subgroup criterion holds. For $n = 1$ this is simply the trivial subgroup.

Exercise 5.49: A positive identity power in a finite group

The powers $e, a, a^2, \ldots, a^{|G|}$ form $|G|+1$ elements counted with repetitions in a set of size $|G|$. Thus $a^i = a^j$ for some $i < j$. Multiplying by a^{-i} gives $a^{j-i} = e$ with positive exponent, as required.

Exercise 5.50: A finite nonempty product-closed subset

Choose $h \in H$. By closure all positive powers of h are in H, and finiteness forces $h^i = h^j$ for some positive $i < j$. In the ambient group cancel to get $h^{j-i} = e \in H$. For any $a \in H$, the same finite-power argument produces $a^m = e$ with $m \geq 1$. If $m = 1$, then $a = e$ and its inverse is in H; otherwise $a^{-1} = a^{m-1} \in H$. Thus H contains the identity and all inverses, and the given product closure completes the subgroup conditions.

Exercise 5.51: The centralizer of one element

The identity commutes with a. If $xa = ax$ and $ya = ay$, then $(xy)a = x(ya) = x(ay) = (xa)y = a(xy)$. To obtain inverse closure, multiply $xa = ax$ on the left and right by x^{-1} to get $ax^{-1} = x^{-1}a$. Thus H_a contains e and is closed under products and inverses, making it a subgroup.

Exercise 5.52: The centralizer of a set and the center

(a) The identity commutes with every $s \in S$. If x, y do so, then for every such s, $(xy)s = x(ys) = x(sy) = (xs)y = s(xy)$. Similarly $xs = sx$ implies $x^{-1}s = sx^{-1}$. Hence H_S is a subgroup. If S is empty the condition is vacuous and $H_S = G$, still a subgroup.

(b) For $S = G$, take $x, y \in H_G$. Since x commutes with *every* element of G, it commutes in particular with y. Thus the subgroup H_G is abelian.

Exercise 5.53: An equivalence relation from a subgroup

Reflexivity follows from $aa^{-1} = e \in H$. If $ab^{-1} \in H$, its inverse ba^{-1} belongs to H, proving symmetry. If $ab^{-1}, bc^{-1} \in H$, their product is $ab^{-1}bc^{-1} = ac^{-1} \in H$, proving transitivity. These are all three required conditions; no commutativity or normality is assumed.

Exercise 5.54: Intersecting two subgroups

Both H and K contain e, so their intersection is nonempty. If $x, y \in H \cap K$, their product lies in H and lies in K, so lies in the intersection. The same applies to x^{-1} because both subgroups are inverse-closed. Associativity is inherited from G, and the subgroup criterion therefore proves $H \cap K \leq G$.

Exercise 5.55: A cyclic group is abelian

Write $G = \langle a \rangle$. Given $x, y \in G$, there are integers r, s with $x = a^r, y = a^s$. Then $xy = a^{r+s} = a^{s+r} = yx$, using commutativity of integer addition, not a preexisting commutativity assumption about G. Thus all elements of G commute.

Exercise 5.56: The set of all nth powers

A sufficient hypothesis is that G is abelian (for any fixed positive integer n). Then $e = e^n$ lies in G_n, the product $a^n b^n = (ab)^n$ is again an nth power, and $(a^n)^{-1} = (a^{-1})^n$ is again an nth power. Thus $G_n \leq G$. The hypothesis is sufficient rather than necessary; when $n = 1$, for example, $G_n = G$ for every group.

Exercise 5.57: No proper nontrivial subgroup forces cyclicity

If $G = \{e\}$ it is cyclic. Otherwise choose $a \neq e$. Its cyclic subgroup $\langle a \rangle$ is nontrivial because it contains a. By hypothesis it cannot be a proper subgroup, so $\langle a \rangle = G$. Thus G is cyclic. Choosing a nonidentity element is the step that excludes the trivial subgroup from this argument.